

NOVA PROGRAM

Nova Network and Diagnostics Requirements

Domain: Communications

Revision A | Synthetic extraction benchmark

Nova Network and Diagnostics Requirements

Legacy controlled requirement sheet

REQ-NVA-NET-001 CAN heartbeat

The controller shall transmit its CAN FD heartbeat at intervals not exceeding 100 ms for a continuous 30-minute operating period.

REQ-NVA-NET-002 Packet loss

During the 30-minute heartbeat test, no heartbeat frame shall be lost at a bus load of 70 percent.

REQ-NVA-NET-003 Ethernet latency

For authenticated diagnostic traffic, round-trip latency over 100BASE-T1 shall remain below 12 ms at 95 percent network utilization.

3. Verification-oriented requirements

Identifier	Requirement title	Normative requirement
REQ-NVA-NET-004	Diagnostic session timeout	An inactive extended diagnostic session shall return to the default session after 5 s, clear the temporary unlock state, and record the timeout event.
REQ-NVA-NET-005	Bus-off recovery	Following bus-off detection, the node shall inhibit transmission for at least 1.0 s, recover without an ignition cycle, and increment the persistent bus-off counter.

Applicability notes

Context	Interpretation
Unless a requirement states otherwise	Limits apply to the complete production-intent assembly.
Use of shall	The complete sentence is normative; table columns do not split obligations.

4. Normative installation figure

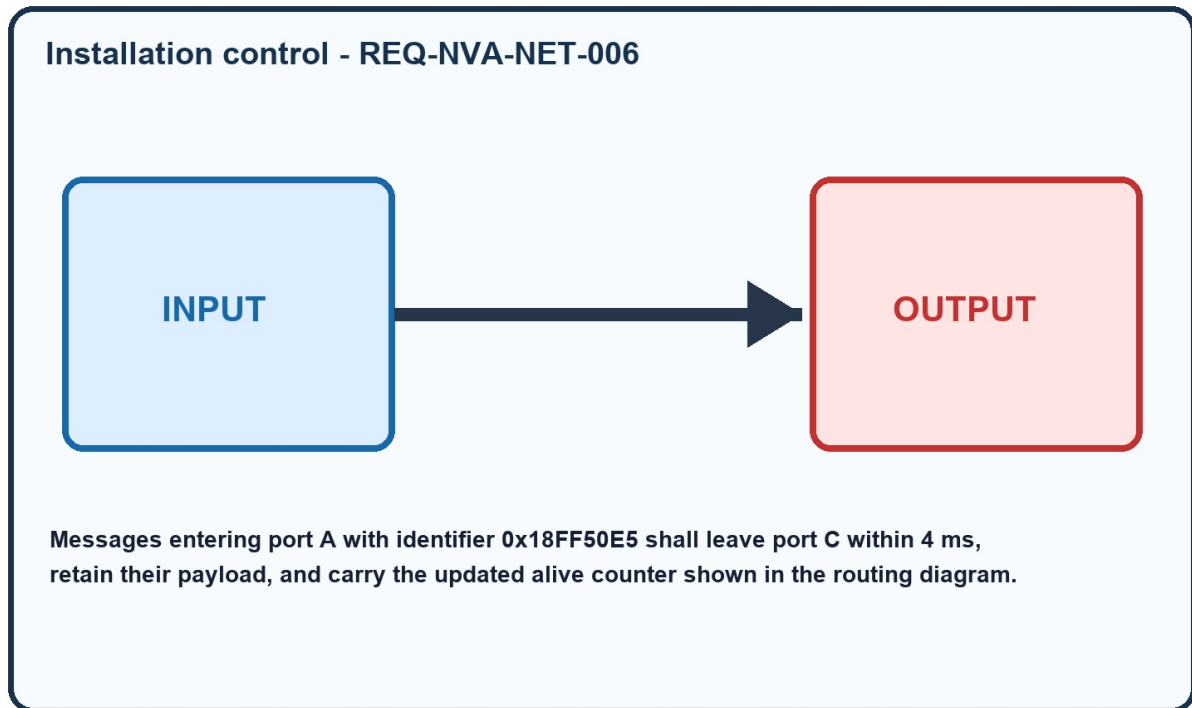


Figure 4-1. Colored geometry and embedded annotation are normative.

Engineering note

The figure annotation forms part of the controlled requirement baseline.

Color, direction, sequence, and geometric relationships must be interpreted visually.